Measured Track-Based Drivetrain Design Framework

Methods, contracts, uncertainty, and decision reporting

Working Phase 8 checkpoint

Framework technical documentation

Version 0.8.0.dev0

Abstract

This document specifies a reproducible method for converting repeated closed-course GPX recordings into drivetrain design evidence. It explains the complete chain: configuration provenance, GPX ingestion, lap reconstruction, centreline and map-matching, physical feature and response-group contracts, speed-gate confidence, the versioned Track Evidence Bundle, vehicle and tire equations, obstacle work, bounded and infinite ideal-CVT mechanisms, gate enforcement, energy closure, uncertainty representation, paired studies, ranking, attribution, hierarchical reporting, and provenance. The document is organized so that the high-level study descriptions in Chapter 13 reference one common set of mechanisms rather than repeating derivations. This is important: a study changes which declared inputs are varied; it does not silently change the physical model.

The current ideal-CVT comparison is one reduced-order mechanism inside the measured track-based framework. Section 18 states the generic interfaces by which external tire, obstacle, or mechanism work may consume the reviewed evidence boundary without rewriting the upstream pipeline. Detailed physical drivetrain-model development is outside this repository's scope.

Contents

Chapter 1

Purpose and scope

1.1 Engineering decision

The framework addresses a decision that is narrower than full lap-time prediction and broader than a single CVT tuning calculation:

> Given reviewed evidence from a measured track, declared vehicle and model uncertainty, and an explicit comparison contract, which tested drivetrain design is preferable, why, and how stable is that conclusion?

The answer has four layers. First, the track evidence must be credible enough to constrain the simulation. Second, the mechanism must close its physical balances and respect measured gates. Third, designs must be compared under paired physical scenarios. Fourth, the report must distinguish numerical validity, physical output variation, finite-sample estimation uncertainty, and directional robustness.

The software does not claim that GPS alone identifies suspension work, tire parameters, driver strategy, or rubber-belt shift mechanics. Unknowns are represented as explicit uncertainty or declared limitations rather than hidden fitting freedom.

1.2 Current flagship comparison

The implemented reference problem compares:

1. a bounded ideal CVT with declared minimum and maximum reduction ratios; and
2. an otherwise identical infinite-ratio reference that removes the operating ratio window but retains finite launch torque.

The infinite case is a counterfactual opportunity bound. It is not a realizable drivetrain and is never reported as a hardware recommendation. Its purpose is to isolate what the finite ratio window prevents under the same engine, efficiency, vehicle, tire, track, gates, driver envelope, and launch-capacity contract. The exact powertrain definition appears in Section 10.2.

1.3 Evidence and model boundaries

The framework keeps three statements separate:

Measured evidence what was observed in GPX passes, feature surveys, video, or another declared source;

Model declaration the equations and parameter distributions used to interpret the evidence; and
Decision synthesis the ranking and warnings produced from completed paired cases.

The Track Evidence Bundle is the boundary between the first two. It contains everything a downstream simulator is allowed to know about the track, but it does not contain the vehicle model or the final decision.

Chapter 2

Architecture and reproducible configuration

2.1 Pipeline

Project and profiles → canonical GPX → reconstructed laps and centreline → reviewed features and gates → Track Evidence Bundle

→ drivetrain/tire/obstacle mechanism → paired study execution → physical and uncertainty attribution → decision-first report

Each arrow is a contract and a provenance checkpoint. The mechanism may consume the bundle, but it may not reinterpret raw GPX internally. Reporting consumes completed machine artifacts and cannot change simulation state.

2.2 Project ownership

A project owns track declarations, run identities, vehicle files, studies, and results. Reusable profiles may provide broad priors or team defaults. Resolution applies layers in a declared order and exports the final merged input. A local override replaces a leaf but does not erase its source history from the resolution manifest.

Every physical scalar is represented as

$$q = (q_0,\ u,\ s,\ \mathcal{D},\ r)\,, \tag{2.1}$$

where q_0 is the nominal value, u is the unit, s is source provenance, $\mathcal{D}$ is an uncertainty declaration, and r is the semantic uncertainty role. A value treated as exact uses a fixed distribution with an explicit reason. This prevents a numerical literal from silently masquerading as perfect knowledge.

2.3 Units and physical support

Values are converted to SI during resolution. Units remain visible in the input contract and human outputs. Distribution support is validated before sampling. The system rejects, for example, nonpositive mass, efficiency above one, a CVT minimum reduction greater than its maximum, negative uncertainty scales, invalid probability vectors, or physically unsupported obstacle parameters. Rejection is preferred to silent clipping because clipping changes the declared distribution.

2.4 Fingerprints

Canonical strict JSON serialization produces a content fingerprint

$$h = \mathrm{SHA256}(\mathrm{canonical}(\mathcal{C}))\,, \tag{2.2}$$

where $\mathcal{C}$ is the resolved scientific contract. Timestamps, process IDs, machine-dependent paths, and cache counters are excluded from scientific fingerprints. Study fingerprints own resume workspaces; simulation fingerprints own cached summaries.

Chapter 3

GPX ingestion

3.1 Run identity

Each GPX file has a unique run identifier plus vehicle and driver identifiers. The manifest independently controls whether the run contributes to centreline geometry and gate evidence. These identities later define a complete paired lap draw in Section 12.4; therefore, distinct sessions must not share one run ID.

3.2 Safe parsing and canonical telemetry

The parser uses a hardened XML implementation. It preserves segment boundaries, timestamps, coordinates, optional elevation, and optional reported speed. Invalid points are diagnosed rather than interpolated invisibly. Time must be monotonic within a usable segment.

If no trustworthy speed field exists, speed is derived from horizontal displacement and elapsed time. For consecutive points $i-1$ and i,

$$v_i = \frac{d_{\text{geo}}(\mathbf{p}_{i-1}, \mathbf{p}_i)}{t_i - t_{i-1}}, \tag{3.1}$$

provided the elapsed time is positive and the interval passes diagnostics. The output records whether each speed was reported or derived. Equation (3.1) is an observation channel, not a vehicle dynamic equation.

3.3 Diagnostics

Ingestion reports missing timestamps, coordinate errors, duplicated points, abnormal time gaps, implausible speed, and derived-speed spikes. Elevation is retained but is not differentiated into road grade in the current release. This limitation is repeated in Chapter 19 because differentiating noisy altitude would introduce a new force model and uncertainty contract.

Chapter 4

Track reconstruction

4.1 Local coordinate frame

Latitude and longitude are projected into a local metric frame (x, y) about a robust course origin. The local approximation is appropriate for a compact endurance course; the original geographic coordinates remain in provenance and review outputs.

4.2 Lap detection

A declared lap-gate event defines $s = 0$. Candidate crossings within a configured radius are ordered in time. Adjacent crossings form candidate laps. A lap is retained only if timing, moving-speed coverage, maximum-speed, data-gap, and later map-matching checks pass. Rejection reasons remain row-level evidence.

This is deliberately not an optimization that selects only the fastest or cleanest laps. It is a quality filter whose thresholds are declared before results are viewed.

4.3 Centreline construction

Valid geometry-contributing laps are aligned to the lap gate, resampled, and combined into a common closed centreline. Let $\mathbf{c}(s) = [x(s), y(s)]^T$ with $s \in [0, L)$. The distance coordinate satisfies

$$s_{j+1} = s_j + \|\mathbf{c}_{j+1} - \mathbf{c}_j\|_2, \qquad L = s_N. \tag{4.1}$$

The output spacing is a configuration choice, not added physical resolution. Median aggregation limits the influence of one poorly aligned pass. A closed-course seam is handled explicitly so features and windows may wrap across start/finish.

4.4 Map matching

Each retained GPX point is projected to the centreline, producing along-track coordinate s, signed/unsigned cross-track error, and local quality information. A lap whose coverage or map error violates the declared threshold is excluded from behavioral evidence. The matched points, not the raw timestamp order alone, define the spatial windows used for gate metrics.

Centreline curvature is calculated for diagnostics. The current longitudinal vehicle model does not spend tire capacity on curvature-derived lateral demand; measured gates represent repeatable corner-entry behavior. This boundary prevents an unvalidated lateral model from being hidden inside track reconstruction.

Chapter 5

Physical features and response groups

5.1 Geometry contract

A feature declares a stable ID, geographic anchor, horizontal uncertainty, interval extent, extent uncertainty, source, analysis role, and optional gate candidacy. After projection, the bundle records start, anchor, end, wrap status, effective position error, and review flags.

The anchor is not automatically the physical entry. The before/after extents express where the declared physical mechanism is active. Entry, approach, exit, and recovery windows are constructed relative to those boundaries.

5.2 Response groups

A response group represents one behavioral speed response. Multiple nearby physical features may belong to one response group if drivers treat them as one compound event. Conversely, separate responses must not be merged only because features are close. One accepted speed gate is attached to a response group, while obstacle energy remains attributable to individual physical features.

This separation prevents double enforcement: a log row and a rut can contribute separate physical work while sharing one measured entry-speed ceiling.

5.3 Pass metrics

For each valid lap and response group, the framework extracts approach speed, entry speed, minimum event speed, braking drop, recovery distance, lap median pace, vehicle, driver, and coordinate quality. The pass table is the auditable basis of the confidence score in Chapter 6 and paired gate sampling in Section 12.4.

Chapter 6

Speed-gate confidence and enforcement

6.1 Confidence components

The gate score is an evidence summary, not a posterior probability. For n eligible passes and target count $n_\star$,

$$c_{\text{count}} = \min\left(1, \frac{n}{n_\star}\right), \tag{6.1}$$

$$c_{\text{repeat}} = \max\left(0, 1 - \frac{\text{IQR}(v_{\text{entry}})}{s_{\text{repeat}}}\right), \tag{6.2}$$

$$c_{\text{brake}} = \frac{1}{n}\sum_{i=1}^{n} \mathbf{1}(\Delta v_{\text{brake},i} \geq \Delta v_\star), \tag{6.3}$$

$$c_{\text{pace}} = 1 - |\text{corr}(v_{\text{entry}}, v_{\text{lap,median}})|, \tag{6.4}$$

$$c_{\text{coord}} = \max\left(0, 1 - \frac{e_{\text{effective}}}{e_{\max}}\right). \tag{6.5}$$

With at least two vehicles, cross-vehicle agreement is

$$c_{\text{vehicle}} = \max\left(0, 1 - \frac{\max_j \widetilde{v}_j - \min_j \widetilde{v}_j}{s_{\text{vehicle}}}\right), \tag{6.6}$$

where $\widetilde{v}_j$ is the vehicle-specific median. A single vehicle receives a declared neutral score of 0.5, not fabricated agreement.

The overall score is

$$C = 100\sum_k w_k c_k, \qquad \sum_k w_k = 1. \tag{6.7}$$

If weights do not sum to one within tolerance, they are normalized and a warning is emitted. The report always exports the components, weights, reasons, and recommendation.

6.2 Recommendations

A candidate with coordinate error above the declared limit is *must fix* regardless of Equation (6.7). Too few valid passes requires review. Otherwise, declared accept and review thresholds classify the gate. The score never overrides a hard geometric failure.

6.3 Braking envelope

An accepted gate at s_g with target v_g is backward propagated using the effective braking deceleration $a_b > 0$. At current coordinate s, the circular distance ahead is $d_g(s)$ and the safe speed is

$$v_{\text{safe},g}(s) = \sqrt{v_g^2 + 2a_b d_g(s)}. \tag{6.8}$$

The driver target is the minimum over all active gates. Effective deceleration is the minimum of the driver acceleration declaration, brake-force capacity divided by mass, and straight-line tire brake capacity divided by mass. A small declared trigger margin starts the full-brake branch before the mathematical boundary to avoid chatter and discrete-step overshoot.

Gate enforcement is one-way: it limits excessive simulated entry speed but never accelerates a weak drivetrain to match a measurement. Compliance is checked at the exact gate crossing and permits only the declared numerical tolerance.

Chapter 7

Track Evidence Bundle

7.1 Purpose

The bundle decouples measured-track reconstruction from drivetrain simulation. It contains the common centreline, reviewed physical features, response groups, accepted gate distributions, paired lap identities, obstacle contracts, capability flags, provenance, schema version, and semantic fingerprint.

It does not contain hidden access to the project, arbitrary Python objects, or an optimized driver. A simulator may therefore replay a reviewed evidence snapshot even when the raw reconstruction pipeline is not rerun.

7.2 Version and checksum

The bundle has a schema version independent of the package version. A consumer accepts only a documented compatible family. Canonical semantic content produces a stable fingerprint; the serialized file has an adjacent SHA-256 checksum. The semantic fingerprint avoids machine-dependent ordering, while the file checksum detects byte changes.

7.3 Capabilities

Capability flags state whether grade force, lateral coupling, gate sampling identities, or other optional evidence exists. Absence is explicit. In the current release, elevation may be stored while grade force remains disabled.

Chapter 8

Obstacle mechanisms

8.1 Spatial work contract

Obstacle models return a local resistance force and optional geometric/traction effects. Energy is accumulated by the line integral

$$E_{\text{obs}} = \int_{s_0}^{s_1} F_{\text{obs}}(s)\, ds. \tag{8.1}$$

The normalized raised-cosine density over an interval of length L is

$$\phi(x; L) = \frac{1 - \cos(2\pi x/L)}{L}, \qquad \int_0^L \phi(x; L)\, dx = 1. \tag{8.2}$$

It distributes a declared event energy smoothly without changing its integral.

8.2 Implemented models

For a fixed specific event energy e_f,

$$F(x) = m e_f\, \phi(x; L). \tag{8.3}$$

For a speed-quadratic event with coefficient k_v and recorded entry speed v_e,

$$E = m e_f + k_v v_e^2, \qquad F(x) = E\, \phi(x; L). \tag{8.4}$$

For distributed roughness with specific energy per distance e_d,

$$F = m e_d. \tag{8.5}$$

The smooth-profile option uses

$$z(x) = \frac{h}{2}\left[1 - \cos\left(\frac{2\pi x}{L}\right)\right], \tag{8.6}$$

$$z'(x) = \frac{h\pi}{L}\sin\left(\frac{2\pi x}{L}\right), \tag{8.7}$$

$$z''(x) = \frac{2\pi^2 h}{L^2}\cos\left(\frac{2\pi x}{L}\right), \tag{8.8}$$

$$\lambda_N = \mathrm{clip}\left(1 + \frac{v^2 z''}{g}, \lambda_{\min}, \lambda_{\max}\right). \tag{8.9}$$

It can alter local slope, normal-load scale, traction multiplier, and unresolved dissipative work. These are reduced-order approximations, not suspension simulation.

8.3 Model-form uncertainty

Alternative obstacle mechanisms are categorical structural uncertainty. A broad coefficient in the bundle is not measured lap variation by default. If repeated event tests support real event-to-event variation, that quantity may be explicitly assigned the measured-track role. The distinction controls which study samples it in Chapter 13.

Chapter 9

Vehicle and tire dynamics

9.1 State

The longitudinal state is distance s, vehicle speed v, and driven-wheel angular speed ω_w:

$$\dot{s} = v. \tag{9.1}$$

Negative vehicle and wheel speeds are excluded by the integration boundary.

9.2 Normal load and tire force

For modeled grade θ, local normal scale λ_N, and driven-load fraction γ, total and driven normal loads are

$$N = mg \max(\cos\theta, 0)\lambda_N, \qquad N_d = \gamma N. \tag{9.2}$$

With surface friction μ, local friction multiplier included in μ, and tire peak scale λ_μ, the force ceiling is

$$F_{\max} = \mu \lambda_\mu N_d. \tag{9.3}$$

Slip speed is

$$v_s = r_w \omega_w - v, \tag{9.4}$$

and the compact saturating tire law is

$$F_t = F_{\max} \tanh\left(\frac{K_s v_s}{F_{\max}}\right). \tag{9.5}$$

The corresponding nonnegative tire-slip dissipation is

$$P_{\text{slip}} = \max(0, F_t v_s). \tag{9.6}$$

Equation (9.5) is deliberately compact. It captures force buildup and a finite ceiling, but not a complete terrain-dependent tire map.

9.3 Resistance and equations of motion

The implemented forces are

$$F_g = mg\sin\theta, \tag{9.7}$$
$$F_{rr} = C_{rr}N\tanh(v/v_0), \tag{9.8}$$
$$F_a = \tfrac{1}{2}\rho C_D A\,v^2, \tag{9.9}$$
$$F_o = \sum_j F_{o,j}. \tag{9.10}$$

The small v_0 regularizes rolling-force direction near rest. Vehicle and wheel dynamics are

$$m\dot{v} = F_t - F_g - F_{rr} - F_a - F_o, \tag{9.11}$$
$$I_w\dot{\omega}_w = T_w - T_b - r_w F_t. \tag{9.12}$$

The model also reports curvature-derived lateral demand $mv^2|\kappa|$ as a diagnostic. It does not reduce $F_{\max}$ because lateral coupling has not yet been validated.

Chapter 10

Powertrain comparison

10.1 Engine

The current engine is a versioned tabulated torque curve scaled by a declared power factor. At engine speed ω_e and throttle $u \in [0, 1]$,

$$T_e = uT_{\text{curve}}(\omega_e), \qquad P_e = T_e\omega_e. \tag{10.1}$$

The ideal CVT targets a declared engine speed $\omega_\star$ associated with target power $P_\star$.

10.2 Bounded ideal CVT

Let final-drive reduction be i_f and CVT reduction be i_c. The ratio required to hold target speed is

$$i_{\text{req}} = \frac{\omega_\star}{i_f\omega_w}. \tag{10.2}$$

The selected bounded ratio is

$$i_c = \text{clip}(i_{\text{req}}, i_{\text{min}}, i_{\text{max}}). \tag{10.3}$$

Inside the window, the engine remains at target speed. Below the minimum-reduction requirement, engine speed rises synchronously with $i_{\text{min}}i_f\omega_w$. Above the maximum-reduction requirement, the declared ideal launch clutch may hold the engine at target while the secondary is slower.

Wheel torque and transmitted power are

$$T_w = \eta i_c i_f T_e, \qquad P_w = \max(0, T_w\omega_w). \tag{10.4}$$

Drivetrain-efficiency loss is $P_e(1 - \eta)$. In launch-clutch mode,

$$P_{\text{clutch}} = \max(0, \eta P_e - P_w). \tag{10.5}$$

Operating shortfall is the counterfactual quantity

$$P_{\text{short}} = \max(0, uP_\star - P_e). \tag{10.6}$$

10.3 Infinite-ratio reference

The reference holds $\omega_e = \omega_\star$ outside a finite operating-ratio window, but it may not produce infinite launch torque. It shares the bounded design's launch cap

$$T_{w,\max}^{\text{launch}} = \eta i_{\max} i_f T_e. \tag{10.7}$$

For nonzero wheel speed, target-power torque is $\eta P_e/\omega_w$. Therefore

$$T_w^\infty = \min\left(\frac{\eta P_e}{\omega_w}, T_{w,\max}^{\text{launch}}\right), \tag{10.8}$$

with the launch cap selected directly at zero speed. Equations (10.3) and (10.8) define the comparison. A design variable may share a cached infinite reference only if it is mathematically absent from this reference mechanism. That proof currently permits a limited minimum-ratio reuse; final drive and maximum ratio are not assumed absent from launch behavior.

Chapter 11

Integration, events, and energy

11.1 Time integration

The solver advances Equations (9.11) and (9.12) with a declared fixed integration step and emits a separate report-step trace. Track completion, maximum time, invalid state, and other termination causes are explicit. Feature-entry and gate-crossing states are interpolated at their exact spatial crossings so a report sampling interval cannot conceal compliance error.

Short coarse-step runs are useful for software-path checks, but they remain invalid if the quality tolerances fail. A production numerical step should be justified by refinement tests on headline metrics, gate excess, and both balances.

11.2 Powertrain energy balance

Integrated engine-side accounting is

$$E_e = E_w + E_\eta + E_{\text{clutch}} + R_p, \tag{11.1}$$

where R_p is the numerical residual. The reported relative residual divides by a finite engine-energy scale.

11.3 Vehicle energy balance

Let total kinetic energy include translation and driven-wheel rotation. Vehicle-side accounting is

$$\begin{aligned} E_w + K_0 = K_f + E_{\text{slip}} + E_{\text{brake}} + E_{rr} + E_a + E_o \\ + W_g + R_v. \end{aligned} \tag{11.2}$$

Grade work W_g is signed. Current GPX elevation does not contribute to it; a smooth declared obstacle may contribute a conservative local grade term. Residual R_v is normalized by transmitted energy, initial kinetic energy, and the magnitude of grade work.

11.4 Opportunity diagnostics

Within one case, the diagnostic

$$E_{\text{opp,total}} = E_{\text{clutch}} + E_{\text{short}} \tag{11.3}$$

captures launch and engine operating opportunity. The bounded-versus-infinite finite-ratio opportunity loss is the nonnegative difference between the two matched cases. It is counterfactual and is *not* added to Equations (11.1) or (11.2).

Chapter 12

Uncertainty representation and sampling

12.1 Semantic roles

The distribution shape and engineering meaning are orthogonal. A stochastic input is assigned one of:

measured track empirical lap/event variability in the observed track;
structural uncertain vehicle, environment, parameter, or model-form assumption; and
initial condition uncertainty in the declared starting state.

This role separation prevents broad obstacle priors from entering measured-track robustness merely because the parameters are stored in the bundle.

12.2 Marginal distributions

Numeric marginals may be fixed, normal, truncated normal, uniform, triangular, or empirical. Categorical model alternatives use declared discrete probabilities. Let $U \sim \mathcal{U}(0,1)$ and F_i^{-1} be the validated inverse marginal transform; then an independent draw is

$$X_i = F_i^{-1}(U_i). \tag{12.1}$$

The nominal value remains the deterministic baseline even if it differs from the distribution median.

12.3 Correlated groups

For a declared Gaussian-copula group with correlation matrix R,

$$\mathbf{Z} \sim \mathcal{N}(\mathbf{0}, R), \tag{12.2}$$

$$U_i = \Phi(Z_i), \tag{12.3}$$

$$X_i = F_i^{-1}(U_i). \tag{12.4}$$

The matrix must be symmetric, positive semidefinite, have unit diagonal, reference known stochastic paths with consistent roles, and not overlap another group. For nonnormal marginals, R describes latent Gaussian dependence; it is not generally the final Pearson correlation.

12.4 Paired gate sampling

When possible, one complete observed run/lap/vehicle/driver identity is drawn and all gate targets use that lap. This preserves within-lap pace relationships. If an otherwise usable identity lacks one gate, only the missing gate may fall back to its empirical distribution, and the gate ID is recorded. Independently combining a fast corner from one lap with a slow obstacle entry from another is therefore not the default.

12.5 Paired designs

For scenario j, all designs receive the same draw $\mathbf{x}_j$. If y_{dj} is an output for design d, a design contrast uses

$$\Delta_{ab,j} = y_{aj} - y_{bj}, \tag{12.5}$$

not a difference between unrelated Monte Carlo samples. Pairing reduces comparison noise and makes scenario-level regret interpretable.

Chapter 13

Study types

This chapter describes what each run changes. It relies on the common track, vehicle, powertrain, gate, energy, and sampling mechanisms defined in Chapters 4–12; no study silently substitutes a different equation.

13.1 Nominal baseline

The baseline uses every declared nominal and the selected nominal gate statistic. It answers whether the bounded and infinite mechanisms execute and what finite ratio range costs in one case. It is a mechanism check, not an uncertainty statement.

13.2 Measured-track robustness

Measured-track robustness samples empirical paired gates and only stochastic inputs with the measured-track role. Structural and initial-condition inputs remain nominal. It answers whether a conclusion survives plausible versions of the measured course.

This study should not be described as "all uncertainty in the track." Geometry uncertainty, grade, and obstacle model uncertainty are included only when the formal contract implements and assigns them appropriately.

13.3 Structural sensitivity

One-at-a-time sensitivity evaluates the exact nominal plus declared marginal quantiles for each selected numeric input. Categorical inputs evaluate the nominal and declared alternatives. Everything else remains nominal.

For numeric response $y(x)$, local slope near the nominal is estimated from the nearest levels bracketing x_0 when possible. A global slope uses all tested levels. The study answers direction and screening magnitude but omits interactions.

13.4 Design sweep

A sweep evaluates explicit values of one design path under common paired scenarios. Design inputs are excluded from random sampling in the same study. It reports both headline metrics, output bands, paired wins, empirical best fractions, regret, threshold probabilities, and convergence. A winning boundary point triggers a recommendation to extend the domain.

13.5 Full uncertainty

Full uncertainty jointly samples every declared stochastic measured-track, structural, and initial-condition input, plus gates under the paired policy. It answers what output distribution follows from all uncertainty the project currently admits. It cannot represent omitted model inadequacy; limitations remain necessary.

Chapter 14

Decision statistics

14.1 Physical output bands

For output samples $y_1, \ldots, y_n$, reports include mean, standard deviation, median, and selected quantiles such as p_{10} and p_{90}. These describe variation across declared physical scenarios.

14.2 Bootstrap estimation intervals

The framework resamples scenarios with replacement and recomputes statistics. Percentile bounds on medians and quantiles describe finite-sample estimation uncertainty. They are not a second physical uncertainty distribution. Paired ranking bootstraps resample whole scenarios so the within-scenario design relationship is retained.

14.3 Wins, best fraction, and regret

For a lower-is-better metric, design d has empirical best indicator

$$B_{dj} = \mathbf{1}\left(y_{dj} = \min_k y_{kj}\right), \tag{14.1}$$

with a documented tie policy. The reported best fraction is the mean of B_{dj} over tested scenarios and designs; it is not a universal probability that the hardware is best outside the declared domain.

Scenario regret is

$$R_{dj} = y_{dj} - \min_k y_{kj} \geq 0. \tag{14.2}$$

Paired win fractions compare two designs on the same scenarios. Threshold fractions report the frequency with which a declared engineering limit is exceeded and include a binomial interval.

14.4 Decision gates

A numerical-quality pass requires all cases to complete, reference dominance, accepted gate compliance, vehicle energy closure, and powertrain energy closure. A directionally robust sweep additionally requires the time and opportunity metrics to select the same winner, acceptable convergence, and paired win-fraction bootstrap lower bounds above 0.5. Numerical validity and directional robustness are separate output fields.

Chapter 15

Attribution

15.1 Physical attribution

Physical attribution reports where energy went using Equations (11.1) and (11.2). It includes design-level bands, scenario-level loss shares, feature-level obstacle work, and closure residuals. Opportunity diagnostics are displayed separately.

15.2 Numeric uncertainty screening

For numeric input x_i and response y, the marginal least-squares slope is

$$\widehat{\beta}_i = \frac{\sum_j (x_{ij} - \bar{x}_i)(y_j - \bar{y})}{\sum_j (x_{ij} - \bar{x}_i)^2}. \tag{15.1}$$

The normalized elasticity at representative values is

$$\mathcal{E}_i = \widehat{\beta}_i \frac{\widetilde{x}_i}{\widetilde{y}}. \tag{15.2}$$

An uncertainty-weighted effect is

$$W_i = |\widehat{\beta}_i| s_{x_i}, \tag{15.3}$$

and relative screening importance is

$$I_i = \frac{W_i^2}{\sum_k W_k^2}. \tag{15.4}$$

Slope bootstrap bounds, Pearson association, and Spearman rank association accompany Equation (15.4). Strong sampled-input correlations generate a warning: marginal slopes are not causal variance partitions.

15.3 Categorical screening

Categorical alternatives report counts and mean output by level, output span, and a between-level separation ratio analogous to eta squared. Sparse levels are flagged. For structural one-at-a-time studies, category values are explicit alternatives rather than pretending that model names have numeric quantiles.

15.4 Sample-count policy

Attribution is suppressed below eight scenarios, exploratory from eight through nineteen, and enabled as screening at twenty or more subject to quality and convergence. This policy prevents a seven-sample smoke test from producing an authoritative-looking tornado plot.

Chapter 16

Convergence and validation

16.1 Monte Carlo convergence

Reports retain scenario count, Monte Carlo standard error where meaningful, and split-half median stability. A status below the screening recommendation explicitly asks for more replicates. The question "are seven samples enough?" is answered by the diagnostics rather than a fixed claim: seven can exercise code but cannot establish a stable distributional decision.

16.2 Numerical validation

The implementation is tested with known limiting cases, exact gate crossings, invalid support, reference dominance, both energy balances, and timestep refinement. Feature obstacle work is reconciled with total obstacle energy. A coarse integration step may complete all code paths yet remain invalid for decision.

16.3 Execution equivalence

Scenario results are sorted by replicate and design after parallel execution. The scientific artifacts must be identical for serial and parallel workers. A content-addressed cache may reduce mechanism-run counts, but cached and uncached scientific outputs must match. Manifest timestamps, worker count, and cache counts are expected operational differences.

16.4 Failure publication

A study writes into a sibling `.incomplete` workspace. Each completed scenario is checkpointed with the owning study fingerprint. On success, reports and provenance are written, checkpoints are removed, and the directory is atomically published. A failed run cannot masquerade as a completed result. Resume rejects changed inputs; restart is explicit.

Chapter 17

Reporting and provenance

17.1 High-to-low hierarchy

Every result is organized as:

1. `SUMMARY.md`: finding or recommendation, confidence, warnings, and next actions;
2. `decision_trace.md`: explicit reasoning chain;
3. `REPORT.md`: compact technical evidence and interpretation;
4. plots: visual location and distribution of effects; and
5. appendix: complete CSV, JSON, JSONL, bundle, resolved inputs, and traces.

The Markdown report is a regenerable projection of machine artifacts. Report regeneration does not rerun physics. Strict JSON disallows nonstandard NaN and infinity tokens; unavailable values use a status and null representation.

17.2 Provenance graph

The provenance record connects measured GPX, track build, exact bundle hash, resolved configuration fingerprint, study fingerprint, framework version, command, and report. The SVG graph is a human index; the JSON is the machine authority. The bundle and resolved inputs make the simulation-facing evidence portable even if original project paths change.

Chapter 18

Extension interfaces

18.1 External mechanism boundary

An external mechanism may consume the same runtime track, driver demand, vehicle state, and paired scenario inputs when it declares a versioned configuration and returns deterministic force, stored-energy, dissipative-loss, and quality channels. The current runner uses its built-in ideal comparison directly; the protocol is an API boundary, not a dynamically selected model system. Any external implementation requires its own validated comparison and caching policy rather than inheriting ideal-reference assumptions without proof.

18.2 Tire and obstacle models

A tire adapter maps tire state and normal load to longitudinal force and physical loss. A brush or empirical model must declare its valid domain, state, sign conventions, energy channel, and uncertainty inputs. Obstacle extensions similarly declare geometry, force/work equations, boundary behavior, physical support, and model-form alternatives.

Upstream GPX and bundle code should not contain conditionals for a particular mechanism family.

Chapter 19

Limitations and future work

19.1 Current limitations

- GPX elevation is preserved but not converted to road grade force.
- The active vehicle model is longitudinal; curvature is diagnostic and lateral/yaw dynamics are absent.
- The tire law is reduced-order and does not resolve terrain-dependent combined slip.
- Obstacle models are explicit approximations and broad defaults are uncalibrated.
- The ideal CVT is an intentionally reduced-order comparison mechanism.
- The driver follows measured gate ceilings and full-throttle/full-brake logic; it is not an optimized human control model.
- Uncertainty attribution is screening, not a global causal decomposition.
- Joint uncertainty cannot represent assumptions that were never declared.

19.2 Prioritized future work

Elevation-to-grade processing requires smoothing, lap alignment, vertical-bias handling, external terrain comparison, and grade confidence. Obstacle calibration requires known geometry, video, vehicle/suspension evidence, and controlled before/after speed tests. Optional expensive global sensitivity may later add Sobol or grouped methods, especially when interactions materially change a drivetrain decision.

Chapter 20

Audit checklist

Before treating a report as engineering evidence, confirm:

1. run identities, timestamps, and speed provenance are credible;
2. lap inclusion and map error were reviewed;
3. physical feature geometry and response groups are correct;
4. accepted gates have credible component scores and enough passes;
5. inherited priors are not described as measurements;
6. uncertainty roles match their engineering meanings;
7. the study type answers the actual question;
8. the design domain includes meaningful alternatives and is extended at a boundary optimum;
9. all numerical quality checks pass;
10. project-validation and track-review evidence blockers are resolved;
11. sample count and convergence support the claimed confidence;
12. energy attribution is not mixed with counterfactual opportunity loss;
13. correlation and sparse-category attribution warnings are respected; and
14. bundle, resolved inputs, seed, fingerprints, and package version are archived.

Appendix A

Study-to-method cross-reference

Run	Varies	Reuses without redefinition
Baseline	No stochastic input	Bundle (Chapter 7), gates (Chapter 6), vehicle (Chapter 9), powertrain (Chapter 10)
Track robustness	Paired gates and measured-track-role inputs	All common mechanisms; structural inputs remain nominal (Chapter 12)
Structural sensitivity	One structural input/alternative at a time	Exact nominal base and common mechanisms (Chapters 12 and 15)
Design sweep	Declared design values under paired scenarios	Pairing, decision metrics, and quality gates (Chapters 12 and 14)
Full uncertainty	All declared stochastic roles jointly	Same physical model and paired gate policy (Chapters 12 and 13)

Appendix B

Output-to-claim cross-reference

Artifact	Claim supported
`SUMMARY.md`	concise finding, confidence, warnings, and next action
`decision_trace.md`	why the headline follows from evidence and gates
`REPORT.md`	technical interpretation, compact tables, and limitations
`replicate_results.csv`	every scenario/design mechanism comparison
`scenario_draws.jsonl`	exact paired inputs and gate identities
`summary.json`	distributions, bootstrap intervals, ranks, regret, thresholds
`energy_accounting.json`	physical partitions and nonadditive diagnostics
`uncertainty_attribution.json`	screening slopes, associations, categories, warnings
`convergence.json`	whether scenario count stabilizes reported metrics
`run_manifest.json`	numerical quality, counts, seed, cache/resume policy
`provenance.json`	versioned evidence and configuration lineage
`track_bundle.json`	exact simulator-facing measured-track contract

Appendix C

Companion documentation

The operational `README.md` gives the shortest path. The companion Markdown files `docs/USER_GUIDE.md`, `docs/INPUT_CONTRACT.md`, `docs/OUTPUT_REFERENCE.md`, `docs/DEVELOPER_GUIDE.md`, and `docs/RELEASE_AND_REPRODUCIBILITY.md` provide field-level and workflow references. Focused contracts under `docs/input`, `docs/output`, and `docs/methods` remain the schema-adjacent source for implementers.